

UNIT

19

ANIMAL BEHAVIOUR





Behavior of Organisms in Response to Stimuli

- Behavior refers to the response of an organism to stimuli.
- It can be inherited but may also be modified by experience.
- Organisms respond to stimuli in various ways, which could be simple actions like moving toward an odor of food or complex behaviors like mating or territory defense.
- Ethology is the study of animal behavior, focusing on how organisms respond to stimuli and their behavioral patterns.

Relationship between Stimuli and Behavior

- **Stimulus:** A detectable physical or chemical change in the environment that triggers a response from an organism.
- **Receptors:** Sensory cells located in various parts of the organism's body that detect environmental changes and convert them into nerve impulses.
- **Interpretation:** The signals received by receptors are sent to the Central Nervous System (CNS) for interpretation. The response can either be:
 - **Reflex Action:** An automatic response that doesn't involve the brain, like a baby jerking away from heat.
 - **Cognitive Response:** A more complex response where the brain interprets the stimulus and the organism reacts accordingly, like a dog salivating at the smell of food

Different Ways of Response to Stimuli

- **Positive Response:** An organism is attracted to the stimulus (e.g., happiness after receiving good news).
- **Negative Response:** An organism avoids or dislikes the stimulus (e.g., a dog hiding when it's time for a bath).
- **Ignored Response:** The organism does not react to the stimulus (e.g., ignoring slight temperature changes in a room).

Relationship between Heredity and Behavior

- The behavior of an organism depends on neural and hormonal mechanisms, including sensory cells, nerve cells, and the CNS.
- **Genes:** Influence behavior by controlling the development of the nervous system and the production of proteins that impact behavior, such as hormones and receptors.
- **Genetic Basis of Behavior:** Behavior can be influenced by genetic variations. For example, selective breeding in dogs has led to differences in behaviors, like hunting skills, among different breeds, even though they belong to the same species.

Biological Rhythms

- **Definition:** Cyclic physiological patterns of activity in an organism, triggered by periodic environmental changes.
- **Biological Clock:** The internal mechanism that controls physiological activities in an organism, independent of external stimuli.





Types of Biological Rhythms

1. Circadian Rhythms (Day–Night Rhythms):

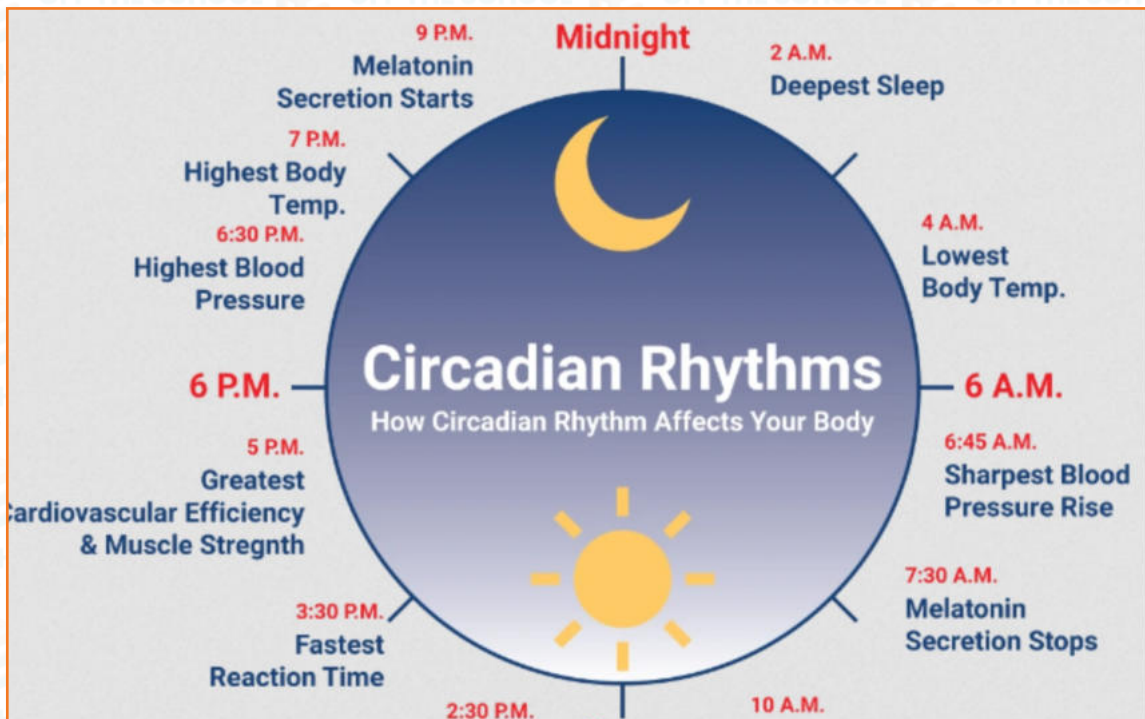
- Rhythms that follow a 24-hour cycle, synchronized with the day–night cycle.
- Organisms adjust their activity patterns based on the time of day (e.g., sleep–wake cycles).

2. Types of Animal Activity Based on Time of Day:

- **Diurnal Animals:** Active during daylight hours (e.g., butterflies, birds, and mammals).
- **Nocturnal Animals:** Active during the night (e.g., cockroaches, owls, and bats).
- **Crepuscular Animals:** Active during dawn or dusk (e.g., mosquitoes).

3. Circannual Rhythms (Annual Cycles):

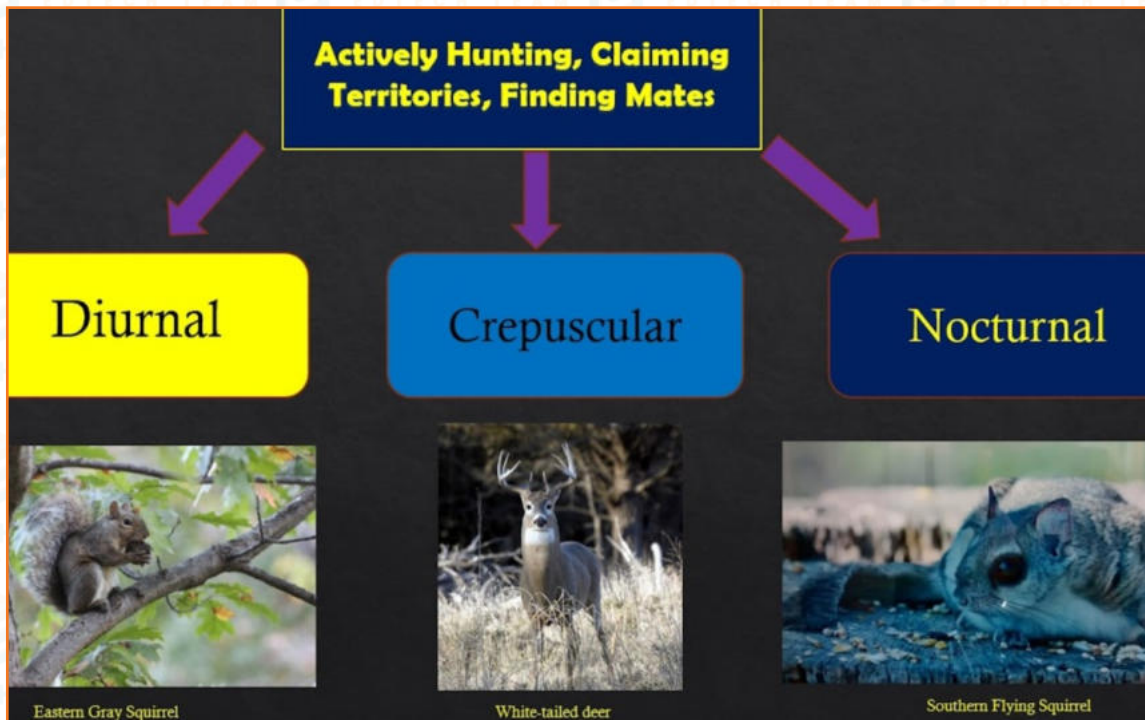
- Rhythms that occur annually, such as:
 - **Bird Migration:** Seasonal migration patterns.
 - **Reproductive Cycles:** Mating and reproductive behaviors in insects, birds, fish, and mammals.
 - **Hibernation:** Seasonal rest or dormancy in some animals.



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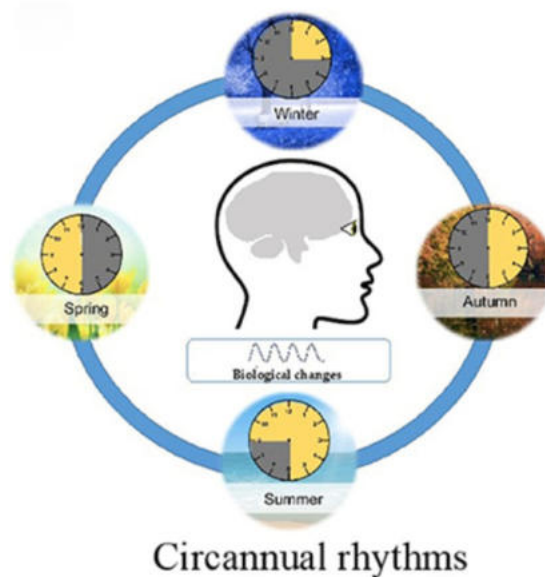
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Types of Animal Behavior

- **Innate Behavior:** Instinctive behaviors that are inherited and do not require learning.
- **Learning Behavior:** Behaviors that are acquired or modified through experience and interaction with the environment.

Innate (Inborn) Behaviour

- **Definition:** Behaviours performed by an organism in response to stimuli at birth, without prior experience. These behaviours are inherited and are generally performed in the same way by all members of a species.
- **Examples:** A human newborn instinctively grasps objects placed in its palm.
- **Importance:** Innate behaviours are crucial for the survival of species, especially those with short lifespans and poorly developed nervous systems.



Spider spinning a web



Bird building a nest



Caterpillar making a cocoon



Dolphin leaping from the water

Types of Innate Behaviours

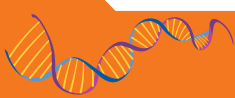
1.Orientation Behaviours:

Behaviours where an organism moves or changes direction in response to a stimulus



2.Non-Oriented Behaviours:

More complex behaviours that do not involve movement toward a specific stimulus. These include reflexes, instincts, and motivations





Orientation Behaviours

- Involve movement toward or away from a stimulus.
- Can be classified into two types: **Taxis** and **Kinesis**.

Taxes

- Definition: A directional movement towards or away from a stimulus.
- **Types of Taxis:**
 - Positive Taxis: Movement towards the stimulus (e.g., moth moving toward light).
 - Negative Taxis: Movement away from the stimulus (e.g., cockroach moving away from light).
- **Types of Stimuli for Taxis:**
 - Phototaxis: Response to light.
 - Chemotaxis: Response to chemicals.
 - Thigmotaxis: Response to touch.
 - Hydrotaxis: Response to moisture.





kineses

- **Definition:** A non-directional movement where the rate of movement changes based on the intensity of the stimulus.
- **Example:** Woodlice moving rapidly in dry environments and slowing down in humid environments



Non-Orientation Behaviours

- More complex than orientation behaviours.
- Involves:
 - **Reflexes:** Automatic, involuntary responses.
 - **Instincts:** Innate, fixed patterns of behaviour.
 - **Motivations:** Internal drives or needs.

Reflexes

- **Definition:** Reflex behaviour is the simplest form of response to stimulation, involving rapid and automatic reactions to simple stimuli.
- **Examples:**
 - **Knee Jerk Reflex:** A stimulus below the knee provokes the spinal cord, which relays it through motor nerves to the extensor muscles, causing a kicking action.
 - **Pupil Constriction:** In bright light, the pupil will contract.
 - **Blinking:** Automatic eye blink in response to an object approaching the eye.
 - **Withdrawal:** Removing your hand from a hot object.
 - **Full Body Involvement:** In some cases, reflexes can involve the entire body, like the withdrawal response in invertebrates (e.g., polychaetes and mollusks). These organisms show rapid, simultaneous contraction of all muscles when triggered by nerve fibers carrying fast signals.

Instincts (Innate or Inborn Behaviour)

Definition: Instinctive behaviour is an inherited and complex pattern of behaviour that does not require learning or practice. These behaviours are genetically programmed and develop with the nervous system. Over generations, these instincts may evolve.

Fixed Action Pattern: Instincts are often referred to as fixed action patterns, where an organism displays a characteristic behaviour without prior experience.





Migration of Salmon Fish

- **Life Cycle:**
 - Salmon spend their life in both freshwater and ocean habitats.
- **Spawning:** Female deposits eggs, and males deposit sperm over the eggs in freshwater.
- **Early Life:** Young salmon stay in freshwater, then leave for the ocean.
- **Ocean Life:** Salmon live in the ocean for several months to years, feeding and growing. After 3 years, they transform into smolt (stage of sexual maturity).
- **Breeding Migration:** In the breeding season, smolts stop feeding and migrate back to the exact stream where they were born for spawning. This journey involves risks like exhaustion or predation.
- Despite the challenges, salmon instinctively complete the journey without prior experience.

The migration of salmon is an excellent example of instinctive behaviour, as they perform this complex task perfectly, even without learning it.



Dances of Bees (Communication in Bees)

- **Purpose:** Honey bees use a unique communication system to convey information about the location and distance of food sources to other bees in the hive.
- **Scout Bees:** A scout bee leaves the hive to search for food. When it returns, it communicates the location of the food to the other bees by performing a special dance.

Types of Dances:

1. Round Dance:

- **When:** Performed if the food source is less than 90 meters away from the hive.
- **Purpose:** Indicates to the other bees that food is nearby, but doesn't provide specific directional information.

2. Waggle Dance:

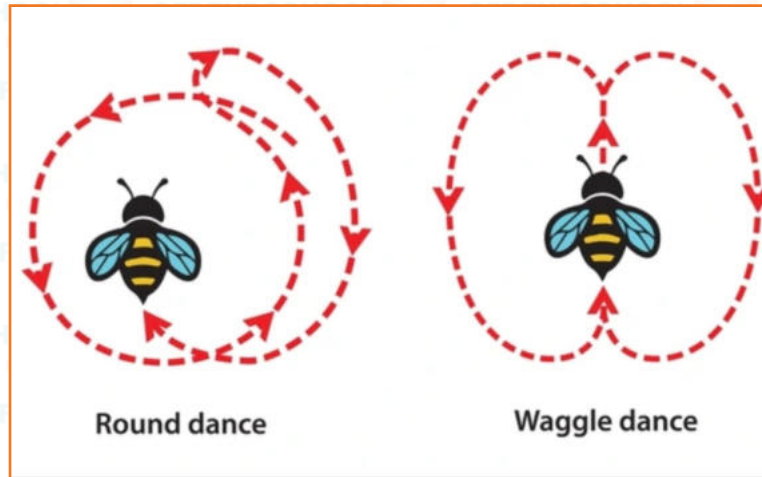
- **When:** Performed if the food source is farther than 90 meters away.
- **Purpose:** Indicates both the direction and distance of the food source from the hive.





How:

- The bee waggles its abdomen and moves in a figure-eight pattern.
- The angle of the waggle dance relative to the vertical in the hive indicates the direction of the food source (relative to the sun).
- The speed of the waggle dance indicates the distance to the food source. A faster dance means the food is farther away, and a slower dance means it is closer.
- Bees can communicate the location of food sources up to 2 kilometers away



Instinctive Behavior:

This dance communication system is an innate or inborn behavior. Bees are born with the ability to perform and understand these dances, enabling them to effectively locate and share food sources.

Instinctive Behavior Examples



Dog shaking after getting wet



Cat arching when scratched



human infant grasps an object





Nest Building by Birds

- **Nest Building Behavior:** All birds have different strategies and techniques for building nests where they lay their eggs. The process of nest building can vary greatly between species.
- **Example:** Long-Tailed Tailor-Bird:
- **Nest Construction:** The long-tailed tailor-bird is known for its ability to build a unique hanging nest. It constructs the nest by weaving twigs and grass between hanging leaves.
- **Instinctive Behavior:** Remarkably, the tailor-bird builds its nest using instinctive behavior rather than learning from older birds. Even without prior experience or instruction, a young tailor-bird instinctively knows how to build its nest.
- **Purpose:** The nest is used for egg laying, and the ability to build this intricate structure is crucial for the bird's reproduction.
- **Instinct:** The long-tailed tailor-bird's nest-building behavior is an innate, inborn instinct that does not require learning from others. This behavior is genetically programmed and develops as part of the bird's natural life cycle.



Building of Spider's Orb Web

Orb Web Construction:

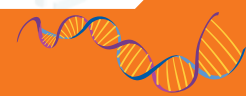
- The orb web is a circular web spun by many common spider species found around the world.
- The web is constructed using silk threads, which are secreted by specialized silk glands located in the spider's abdomen.
- The silk is a scleroprotein secretion, which is initially emitted as a liquid. It hardens upon contact with air, forming the durable silk threads.

Instinctive Behavior:

- The spider follows a series of basic rules or patterns, such as measuring angles and walking certain distances along threads.
- By adhering to these elementary rules, the spider is able to construct a complex structure without needing a detailed plan in its head.

Complexity and Instinct:

- The ability to build the orb web is not learned but is an instinctual behavior, genetically programmed into the spider.
- This behavior is highly complex and results in a perfectly functional web, despite the spider having no conscious understanding of the structure it is creating.





Purpose of the Web:

- The orb web serves various functions, such as trapping prey and providing a safe environment for the spider to live and reproduce.



Courtship Behavior of Stickleback Fish

Nuptial Coloration:

- The three-spined stickleback fish exhibits ritualized courtship and aggressive behavior.
- During the breeding season, sexually mature males develop bright nuptial coloration, characterized by a red belly and a brightly colored throat. This coloration is a signal to females that the male is ready to mate.

Territorial Behavior:

- The territorial male stickleback builds a nest to attract a female and protect his breeding territory.
- He becomes aggressive and dominant, defending his nest from other males.

Courtship Ritual:

- A female with a prominent silver belly approaches the male's territory.
- The male leads the female to the nest, demonstrating the entrance by moving around it and encouraging her to enter.

Spawning Process:

- Once the female enters the nest, the male stimulates spawning by gently nudging the base of her tail.
- After spawning, the female leaves, and the male stays behind to fertilize the eggs inside the nest.

Parental Care:

- The male stays close to the nest, fanning the eggs with his fins to ensure they receive oxygen.
- He protects the newly hatched fish for several days, providing care and ensuring their safety from predators.

Instinctive Behavior:

The entire courtship and parental behavior of the stickleback fish is instinctive—it is genetically programmed and does not require prior experience for successful execution.





Learning Behavior

- **Definition:**
Learning is a process through which animals modify their behavior based on specific experiences. This modification is **adaptive**, as it helps the animal adjust to its environment and improve responses to similar situations in the future.
- **Key Points:**
 - Unlike innate behavior, **learning behavior** is not controlled by genes but is shaped by **experiences** from the environment.
 - **Learning** is more prominent in animals with **longer lifespans** and a **well-developed nervous system**, which allows them to gain knowledge and adapt over time.
 - Learning allows animals to **adapt to changes** in their environment, improving survival and response to challenges.

Distinguishing Between Innate and Learning Behavior

INNATE BEHAVIER	LEARNING BEHAVIOR
Inherited and present from birth.	Acquired through knowledge or experience.
Reflex actions when exposed to a stimulus.	Learned or acquired through experience.
Permanent and remains the same across generations .	Can be modified by experience and may vary between individuals.
Contributes to the survival and proper functioning of organisms.	Helps animals improve behavioral traits to fit into their environment.
Common in animals with short lifespans .	Common in animals with long lifespans and developed nervous systems .
Requires no time to adapt.	Requires more time to adapt and learn.
Example: A newborn baby kicking reflex .	Example: Learning to play cricket through practice.





Habituation

• Definition

Habituation is the simplest form of learning in which an animal **learns not to respond** or **ignores** a repeated, **irrelevant stimulus**. Over time, the animal becomes accustomed to a harmless stimulus and no longer reacts to it.

• Key Points

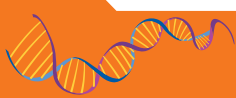
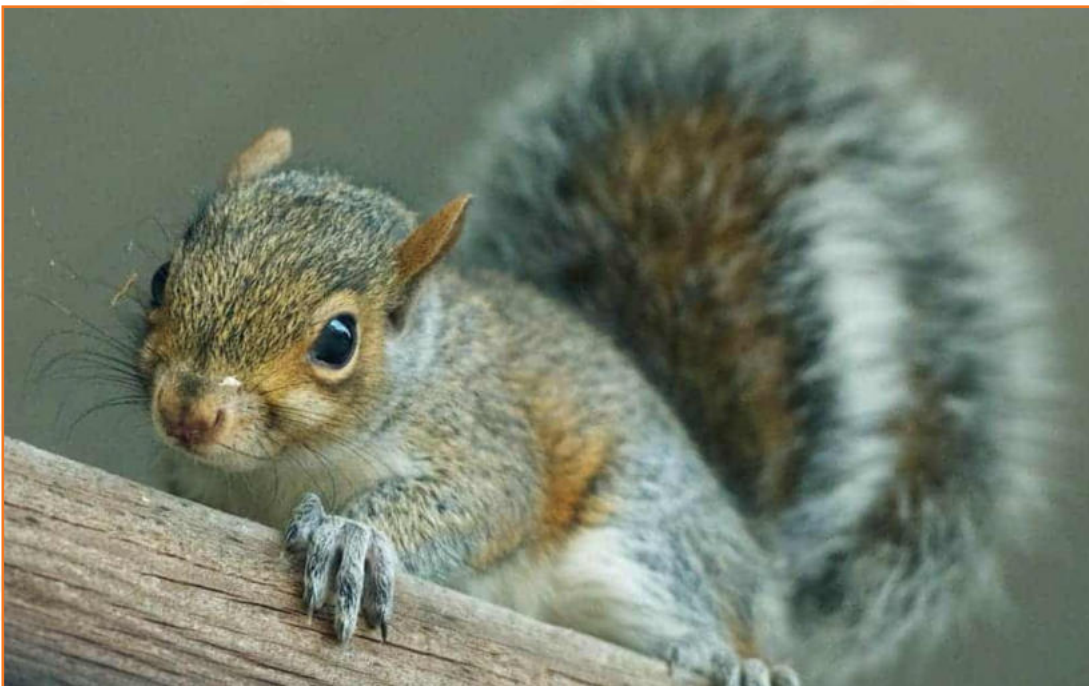
- The animal **perceives the stimulus**, but **decides** to ignore it after repeated exposure.
- It occurs at the level of the **brain** and involves **processing** stimuli without triggering a response.
- **Stimuli may still be detected**, but the animal's **behavioral response diminishes** with repeated exposure.

• Example:

- Wild squirrels in a park initially **move away** when a person approaches because they associate humans with danger. Over time, as they experience repeated harmless interactions, the squirrels **learn** that humans are not a threat.
- Eventually, they may approach humans for food, **losing their fear** due to habituation.

• Real-World Application:

Habituation helps animals **save energy** by not reacting to every insignificant or non-threatening stimulus in their environment, allowing them to focus on more relevant or dangerous stimuli.





Imprinting

● Definition:

Imprinting is a type of learning where a young animal **fixes its attention** on the first **moving object** or another animal it encounters shortly after birth or hatching. This learning typically occurs during a **short sensitive period** early in life.

● Key Points:

- Once an animal has imprinted, the learned behavior is **firmly fixed** and may influence later life experiences.
- Imprinting helps young animals with **social interactions** such as identifying mates, forming flocks, or bonding with parents.
- This form of learning is **irreversible** after the sensitive period.

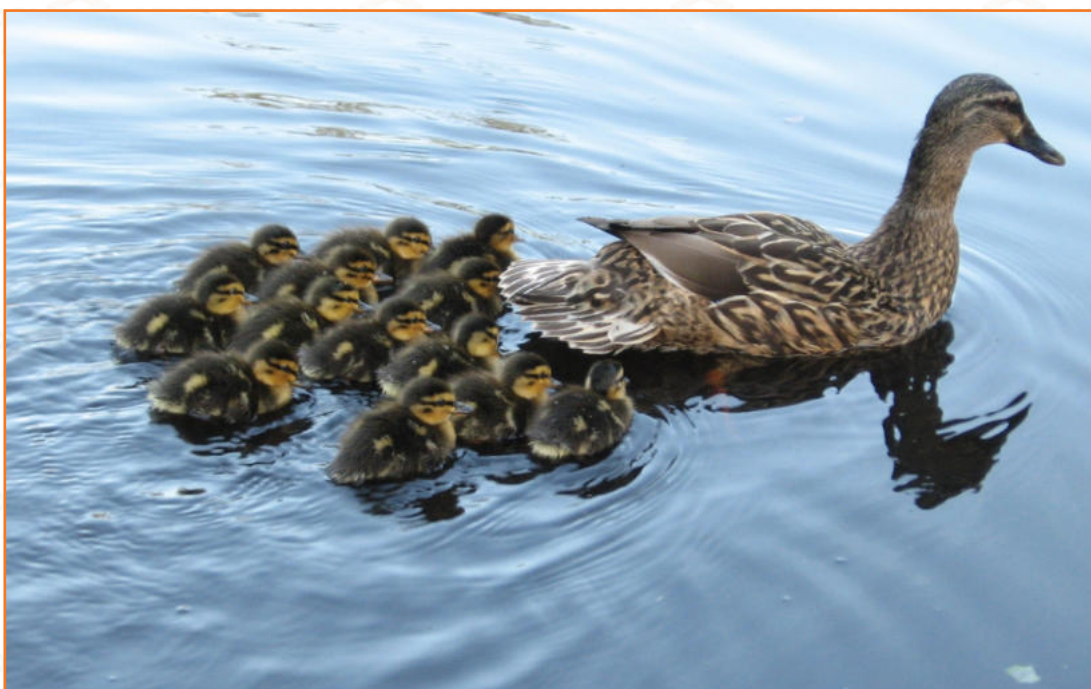
● Classic Example:

Konrad Lorenz (1930s) famously demonstrated imprinting using **greylag geese** (*Anser anser*).

- Lorenz found that goslings would **follow the first moving object** they saw after hatching. By substituting himself for the mother, he could make the goslings imprint on him. They followed him instead of their mother, even as adults.
- **Key Observation:** The primary imprinting stimulus was **movement** of an object (typically the mother).

● Significance:

Imprinting is a crucial **adaptive mechanism** for young animals to learn vital behaviors, such as bonding with their parents and recognizing social structures.





Difference Between Habituation and Imprinting as Reversible and Irreversible Learned Behavior

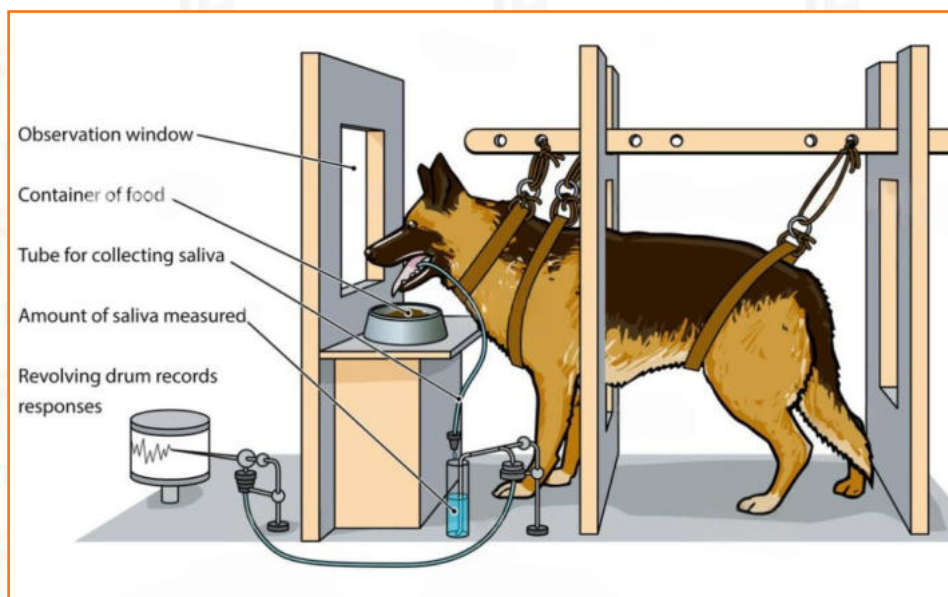
ASPECT	HABITUATION	IMPRINTING
Definition	A form of learning where an animal stops responding to a repeated, irrelevant stimulus	A type of learning where an animal fixates its attention on the first moving object it sees early in life.
Reversibility	Reversible: After a pause, the animal may start responding again to the stimulus.	Irreversible: Once the behavior is learned, it remains throughout life .
Example	A snail retracts into its shell when tapped on a glass, but eventually stops reacting to the tapping.	A young bird follows the first moving object it sees after hatching, and continues to follow it even into adulthood.
Timeframe	Takes place over short periods of time ; the response gradually diminishes.	Takes place during a critical sensitive period early in life.
Nature of Learning	A gradual decrease in response to a stimulus due to repeated exposure.	A firm and lasting attachment or behavior towards the stimulus.
Impact on Behavior	The animal resumes normal behavior after a break from the stimulus.	The learned behavior remains fixed and influences social or mating behavior for life.





Classical Conditioning

- **Definition:** Classical conditioning is a form of learning where an animal **learns to associate two different stimuli**, leading to a response that would not naturally be triggered by one of the stimuli.
- **Famous Experiment:**
 - Developed by **Ivan Petrovich Pavlov**.
 - Pavlov's experiment involved ringing a bell simultaneously with presenting food to a dog.
 - After repeated pairings, the dog began to **salivate** in response to the bell alone, even when no food was presented.
- **Key Terms:**
 - **Unconditioned Stimulus (UCS):** A stimulus that naturally triggers a response (e.g., **food** triggers salivation).
 - **Conditioned Stimulus (CS):** A previously neutral stimulus that, after association with the UCS, triggers a similar response (e.g., **bell** becomes associated with food and causes salivation).
 - **Unconditioned Response (UCR):** The natural response to the unconditioned stimulus (e.g., **salivation** due to food).
 - **Conditioned Response (CR):** The learned response to the conditioned stimulus (e.g., **salivation** due to the bell alone).
- **Process:**
 - **Neutral Stimulus (NS)** (bell) is presented alongside the UCS (food).
 - Repeated pairing of the NS and UCS causes the animal to associate them.
 - Over time, the NS becomes a CS, triggering the CR (salivation) even without the UCS (food).
- **Example:**
 - Initially, **food (UCS)** triggers salivation (**UCR**).





- After repeated pairings of **bell (NS)** with **food (UCS)**, the bell alone causes the dog to salivate (**CR**).

Instrumental Conditioning (Trial and Error Learning)

- **Definition:** Instrumental conditioning, also known as **operant learning**, involves learning through **trial and error**. The animal learns by performing certain actions, and when these actions lead to a rewarding outcome (like food), the animal associates the behavior with the reward.

- **Key Process:**

1. The animal engages in **searching actions** (e.g., walking, moving around).
2. The animal **accidentally or eventually** performs an action (e.g., pressing a lever) that results in a **reward** (e.g., food).
3. The **reward reinforces** the behavior, and the animal learns to associate the action with the positive outcome.
4. Over time, the animal **repeats** the behavior to obtain the reward.

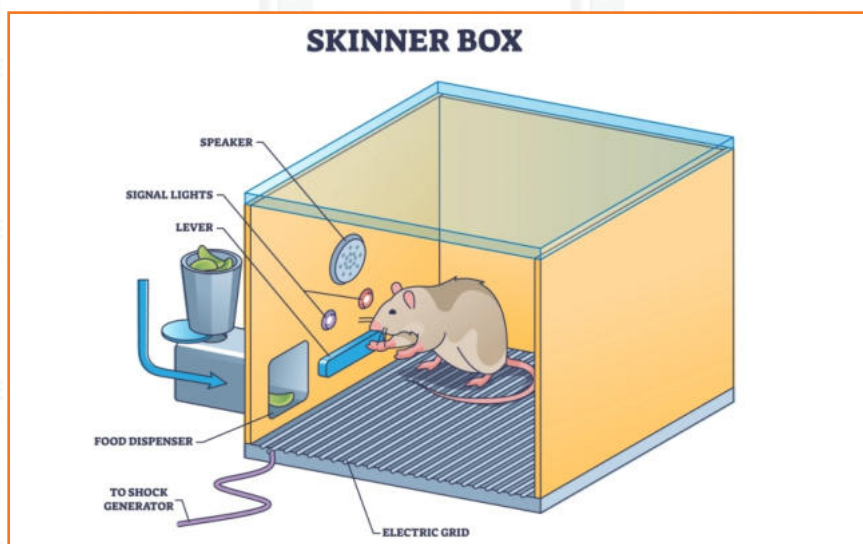
- **Example:**

1. **B.F. Skinner's** famous experiment with a **Skinner box**:

- A **rat** is placed in the box.
- The rat explores the box, moving around.
- Eventually, the rat **presses a lever** by accident.
- The rat receives a **food pellet** as a reward.
- Through repeated trials, the rat learns that pressing the lever leads to food.
- The rat will eventually **press the lever immediately** to obtain food.

- **Key Concept:**

1. **Reinforcement:** The reward (food) strengthens the behavior, making it more likely to occur again.
2. The animal is **instrumental** in providing its own reinforcement by performing the desired action.

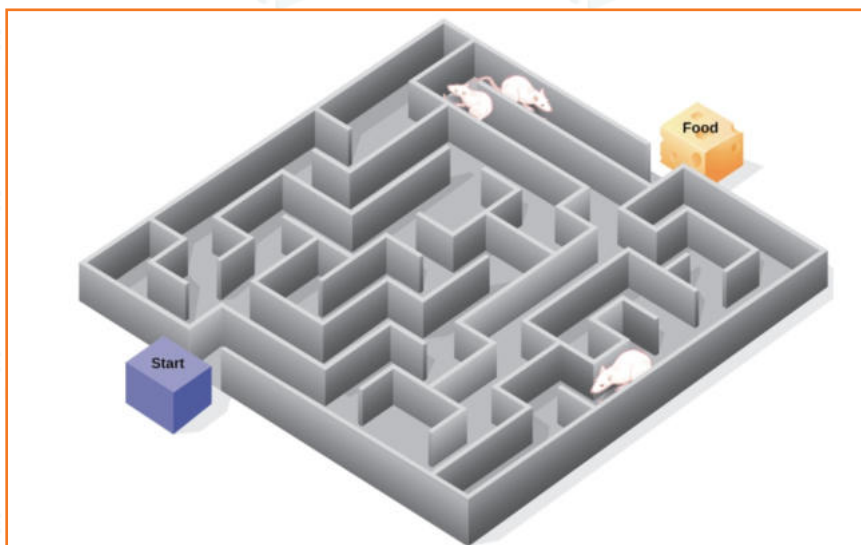




Latent Learning

- **Definition: Latent learning** is a type of learning that occurs **without immediate reinforcement or motivation**. In this type of learning, animals may explore or experience their environment, but the knowledge or learning is not immediately demonstrated until there is a reason or motivation to use it (e.g., when food or a reward is present).
- **Key Concept:**
 - **Learning without reinforcement:** Animals can gather information and learn about their environment without being reinforced or rewarded at that time. This knowledge becomes useful later when there is a **motivation** to apply it (e.g., when the animal is hungry).
- **Example (K.L. Lashley's Maze Experiment):**
 - A **rat** is placed in a **maze** and explores it to find the exit.
 - While exploring, the rat **learns** the layout of the maze and the location of food, but it **does not need** the food at that moment.
 - Later, the same rat is placed in the maze again, but this time it is **hungry**.
 - The rat quickly finds the **food**, using the **knowledge it had gained earlier** during the first exploration.
- **Key Concept:**
 - **Latent learning** does not require immediate reinforcement. The rat had learned the maze layout earlier but did not demonstrate the learning until it had a **motive** (hunger) to apply it.

This type of learning suggests that animals (and even humans) can acquire information in an environment, and the learning may not be immediately observable until it is required.

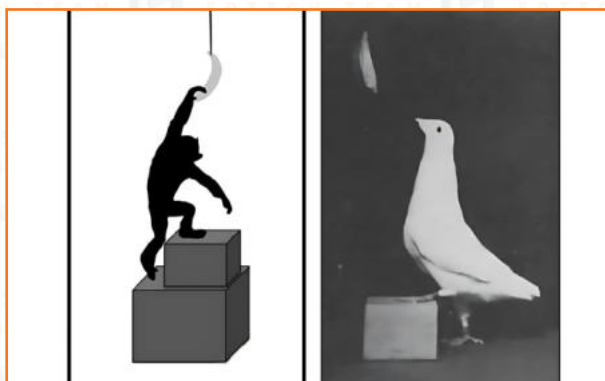




Insight Learning

- **Definition: Insight learning** is a form of learning where animals use **cognitive processes** (like thinking, understanding, or problem-solving) to solve problems without prior experience. It involves a **sudden realization** of how to solve a problem, often after considering various possibilities or observing the environment.
- **Key Concept:**
 - **Cognitive problem-solving:** Insight learning is distinct because it involves a **cognitive breakthrough** rather than relying purely on trial and error. The animal **actively thinks** and **analyzes** the situation to come up with a solution.
 - **Example (Wolfgang Köhler's Experiment with Chimpanzees):**
 - **Setup:** German psychologist Wolfgang Köhler conducted an experiment in the 1920s on **chimpanzees**. He placed chimpanzees in a **cage** with **bananas hanging out of reach**, and provided them with **several sticks** and **boxes**.
 - **Initial Attempt:** The chimpanzees first tried to **reach** the bananas but failed, as they were too high up.
 - **Insightful Solution:** After a moment of observation and thinking, the chimpanzees used their cognitive skills to solve the problem:
 - They **stacked** the boxes on top of each other to **reach** the bananas.
 - Alternatively, they **combined sticks** to create a longer tool that could knock the bananas down.
 - **Conclusion:** Köhler called this **insight learning** because the chimpanzees didn't learn by trial and error. Instead, they **thought** about the problem and came up with a creative solution.
 - **Key Concept:**
 - **Insight learning** involves **problem-solving** through mental processes rather than trial and error, making it a form of **higher-level cognition**.

This type of learning shows that animals, especially those with more developed cognitive abilities like primates, can engage in **complex thinking** to solve problems in ways that don't rely on experience or repetitive trials.





Social Learning

- **Definition:** Social learning refers to the process by which animals learn behaviors through interaction with others, rather than through direct personal experience. It involves the observation, imitation, and modification of behaviors based on the actions of others.



Aggregation and Animal Societies

1. Aggregation :

- **Definition:** Aggregation is a simple gathering of animals that are in close proximity to each other but do not interact socially or cooperate. They may be grouped together for basic needs like feeding, drinking, or mating, but the members of the group do not necessarily engage in direct interaction.
- **Examples:**
 - **Fruit flies** hovering around rotting fruit.
 - **Flocks of birds** of various species living together in trees.
 - **Groups of zebras** or a **school of fish**.
- **Characteristics:** These groups may be temporary, and the animals do not exhibit complex social behaviors.





2. Animal Societies:

- **Definition:** An **animal society** or **social group** is a group of animals of the same species that live together and exhibit cooperative social relationships. These societies are characterized by complex, organized behaviors and cooperation among members.
- **Characteristics:**
 - **Cooperative relationships:** Members of the group engage in activities such as **feeding, water-sharing, body care, and sexual interactions.**
 - **Mutual attraction:** The individuals in the group are often drawn together by mutual attraction and may engage in behaviors that support the group's survival and functioning.
 - **Social bonds:** The social group can be considered a **permanent union** of individuals who interact regularly for the benefit of all.



Hostile and Helpful Intraspecific Interactions

- **Intraspecific Interaction:** Interaction between individuals of the same species.

Hostile Interaction:

- **Definition:** In hostile interactions, individuals of the same species may engage in aggressive or harmful behaviors toward each other.
- **Example in Honey Bees:**
 - As **worker bees** age and become less efficient at performing their duties, they are often killed by other workers. This ensures the hive's efficiency by eliminating those no longer able to contribute productively.
 - This can be seen as a **self-regulating mechanism** within the colony, ensuring that only the most capable individuals remain to perform essential tasks.





Helpful Interaction:

- **Definition:** In helpful interactions, individuals cooperate for mutual benefit, often working together to achieve common goals.
- **Example in Honey Bees:**
 - **Workers:** Collect nectar and transform it into honey, contributing to the hive's food reserves.
 - **Drones:** Their primary role is to **fertilize the queen's eggs** and **provide defense mechanisms** for the hive.
 - **Queen:** Lays eggs that develop into new workers, drones, or new queens, maintaining the population and ensuring the colony's survival.
 - All castes in the hive work cooperatively to sustain the colony's overall health, reproduction, and defense.

Agonistic Behaviour

- **Definition:** Agonistic behaviour refers to the range of aggressive or conflict-related interactions between animals, especially over access to limited resources such as food, mates, or territory. This behaviour helps animals resolve disputes and establish dominance or territorial control.

Key Features of Agonistic Behaviour:

1. **Aggression:** One animal becomes aggressive or attacks another to gain access to resources.
2. **Threats:** Often, animals issue threats to warn competitors before resorting to actual aggression or physical combat.
3. **Rituals:** In many species, agonistic behaviour is expressed through ritualized displays or behaviors that signal dominance or submission without physical confrontation.
4. **Combat:** In some cases, agonistic behaviour escalates to physical combat to determine which competitor gains control of the resource.

Significance of Agonistic Behaviour:

- **Territory Maintenance:** Agonistic behaviours help animals defend their territories from intruders. This is especially important for species that rely on a specific area for resources like food or nesting sites.
- **Dominance Hierarchies:** In social groups, agonistic behaviour establishes dominance hierarchies, with some individuals asserting dominance over others. This hierarchy helps reduce conflicts by providing a clear social order and reducing the need for constant aggressive encounters.
- **Resource Access:** Agonistic interactions determine which animal gets access to resources such as mates or food. For example, male animals often display aggressive behaviour to defend their territory or attract mates.





Examples:

- **Males in Many Species:** Male animals often show aggressive signals or engage in physical combat to establish their right to mate or defend their territory.
- **Birds and Mammals:** In birds, males may engage in song battles or physical displays (like pecking) to establish territory, while male mammals might fight with each other for dominance or access to females.



Territory in Animals

- **Definition:** A territory is a fixed area that an animal actively defends against others of the same species. It is typically used for essential activities such as feeding, mating, and rearing young.

Characteristics of Territorial Behaviour:

1. **Territorial Defense:** Territorial animals use **agonistic behaviour** (aggression, threats, and rituals) to defend their territory against intruders.
2. **Functions of Territory:**
 - **Feeding:** Ensures access to a reliable food source.
 - **Mating:** Male animals often defend territory to attract and court females without interference.
 - **Rearing Young:** Provides a safe space for raising offspring.
3. **Territorial Marking:** Animals often mark their territory using scent markings, vocalizations, or physical displays to warn others of their claim.





Territorial Behaviour in Gorillas:

- **Social Structure:** Gorillas live in social groups called **troops**, consisting of one or more dominant males (silverbacks), several females, and their offspring.
- **Dominant Male:** The adult silverback is the dominant member of the troop and has exclusive breeding rights to the females. He is also responsible for the safety and protection of the troop.
- **Territoriality in Gorillas:**
 - While gorillas are **non-territorial**, the dominant silverback exhibits some territorial behaviour, especially in defending his troop.
 - The silverback may stand upright and beat his chest to intimidate threats, displaying aggression to defend his family.
- **Family Structure:**
 - Females join the troop and form relationships based on rank, which is determined by their time of arrival. Later arrivals have lower ranks.
 - Adolescent males, upon reaching a certain age, leave the troop to avoid competition for mates and often remain solitary until they form their own troop.



General Observations of Territorial Behaviour:

- Territorial behaviour is seen across many species, including **worms, insects, fish, birds, and mammals**.
- In many species, **territoriality** helps ensure reproductive success, food security, and safety from predators.





Territorial Behaviour in Baboons

• Species Overview:

- Baboons are **Old World monkeys**, characterized by non-prehensile tails and downward-facing nostrils. They are part of the genus *Papio* and are ground-dwelling primates.
- Baboons live in **troops** within a defined **territory** and are known for their social structure and complex behaviours.

Key Features of Baboons' Territorial Behaviour:

1. Sexual Dimorphism:

- Male baboons are **larger** than females, and this size difference plays a role in social interactions and dominance.

2. Dominance Hierarchy:

- **Dominant Males:** One or more male baboons establish dominance over other males by intimidating them, often using their long, sharp **canine teeth**.
- **Costs of Dominance:** The dominant males take on the responsibility of defending the troop from predators, which can result in injuries during confrontations.

3. Territorial Movement:

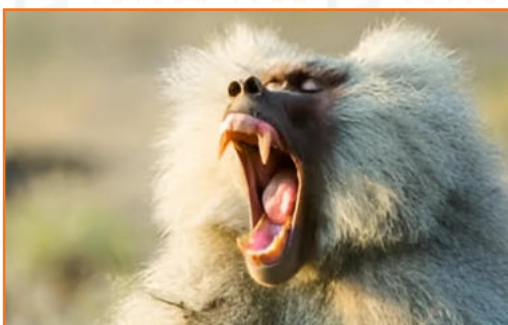
- Baboons **forage** throughout their territory during the day and sleep in **trees** at night for safety.
- **Dominant males** decide where and when the troop will move, influencing the troop's daily activities.

4. Defense Mechanisms:

- **Protection:** Dominant males are responsible for defending the troop from external threats.
- **Vocalization and Displays:** Baboons often rely on **vocalizations** and **displays** (rather than physical fighting) to defend their territory and scare off intruders.

5. Cooperative Defence:

- In cases of danger, the **dominant males** lead the troop in retreat and take an active role in attacking or defending against intruders when necessary.





Social Structure:

- Baboons live in complex **social hierarchies** within their troops, and the dominant male plays a crucial role in maintaining order and ensuring the survival of the troop. Their territory is an essential part of their social and survival strategy.

Dominance Hierarchy

- **Definition:** A **dominance hierarchy** is a social ranking system where animals are organized based on their rank, determining access to resources like food, mating opportunities, and grooming from other members of the group.
- **Top-ranked Individuals:** Those at the top of the hierarchy typically have **first access** to food, more mating opportunities, and safer territories compared to those at lower ranks.

Characteristics of Dominance Hierarchies:

1. Aggressive Interactions:

- The hierarchy is often established through **aggressive behaviours** and **responses** to these behaviours among individuals within a group.
- Over time, these interactions settle into a **stable ranking** where each animal knows its position in the social structure.

2. Benefits of High Rank:

- Higher-ranked individuals enjoy better access to food, mates, and territory, giving them evolutionary advantages.
- **Alpha individuals** tend to have more control over resources and exhibit fewer direct confrontations because their dominance is well established.

Example: Pecking Order in Chickens

● Establishment of Pecking Order:

- When unfamiliar hens are placed together, they initially engage in **fighting, chasing, and pecking** to establish a **clear pecking order**.
- This hierarchy determines which hen has dominance over the others in terms of access to resources.

● Alpha Hen:

- The **alpha hen** (top-ranked) is the most dominant and is not pecked by other hens.
- The alpha hen can **drive off** others through threats rather than physical pecking.
- She has **first access** to food, water, and roosting sites.





Beta Hen:

- The **beta hen** (second-ranked) subdues all hens except the alpha, holding a secondary position in the hierarchy.

Omega Hen:

- The **omega hen** (lowest-ranked) is the most submissive, often avoiding confrontation and getting the least access to resources.



Hierarchy Stability:

- Once the pecking order is established, peaceful coexistence is often possible.
- **Occasional conflicts** can arise if an individual attempts to move up in the social ranking.

Importance of Dominance Hierarchies:

- **Resource Allocation:** Dominance hierarchies are critical for determining how limited resources are distributed among group members.
- **Social Order:** The hierarchy helps in maintaining **social stability** and **order** within animal groups, reducing constant physical conflict and improving survival chances.

Altruism (L. alter, the other)

- **Definition:** Altruism is a behavior performed by an individual that benefits others in its society, even at a cost to its own survival or reproductive success. The animal sacrifices some of its own potential to benefit others, often seen in social animals.
- **Contrast with Selfish Behavior:** While many behaviors are selfish (focused on the individual's own benefits), altruism is selfless, with no direct benefit to the individual performing the act.
- **Significance:** Altruistic behavior is common in social species, where individuals cooperate for the benefit of the group.





Altruism in Honeybee Societies

- **Social Structure:** Honeybee colonies consist of three castes, each with distinct roles:

1. **Queen:** The sole fertile female responsible for laying eggs (both fertilized and unfertilized).
2. **Workers:** Sterile females that perform all the labor within the colony, such as:
 - Cleaning and polishing the hive
 - Building beeswax combs
 - Foraging for nectar
 - Caring for the queen
 - Guarding the hive entrance
3. **Drones:** Male bees that develop from unfertilized eggs. Their primary role is to mate with the queen, after which they die.

Reproductive Role:

1. The **queen** produces fertilized eggs that become workers or new queens, and unfertilized eggs that become drones.
2. The **worker bees**, despite being sterile, dedicate their lives to supporting the colony, protecting the queen, and ensuring the colony's survival.

Altruistic Behavior in Honeybees:

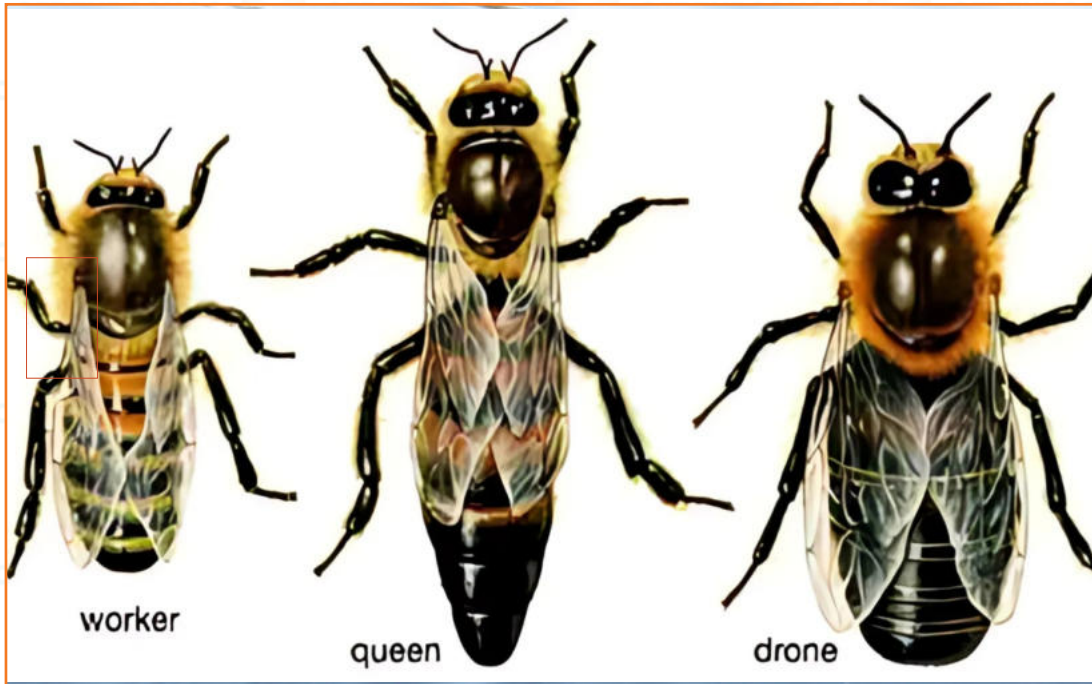
1. **Worker bees** exhibit altruistic behavior by sacrificing their ability to reproduce in favor of supporting the colony. Their primary function is to help maintain and protect the hive, ensuring the success of the queen's offspring.
2. In some cases, when a new queen emerges, **worker bees** may sacrifice the new queen to prevent a challenge to the established reproductive system of the colony.





Colony Dynamics:

1. When a new queen is produced, the colony may split, with the old queen with a group of workers and drones to establish a new hive, leaving behind the newly emerged queen.
2. **Drones**, whose sole purpose is to mate with the queen, die after mating, a form of self-sacrifice for the colony's reproductive success.



Key Takeaways:

- **Altruism** benefits the group or colony, even at the cost of the individual's reproductive success.
- In honeybees, **worker bees** engage in altruistic behavior, dedicating their lives to maintaining and supporting the hive, despite being sterile.
- **Drones** and **worker bees** are examples of how some individuals sacrifice their own well-being for the benefit of the colony's survival and reproductive success.

